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BTech CSE - A.

Assignment → 2

Ans:1 → 1) Entity relationship Diagram
2) Relational
3) Object - oriented
4) Integrity.

Ans:2 → Data abstraction means hiding complex databases details and showing only important information to users. In a healthcare system, it helps doctors, nurses and staff access patients records easily. There are three levels of data abstraction :-

1) Physical level :- Describes how data is stored in the database. Users do not interact with this level.
2) Logical level :- Defines what data is stored, such as patient details, doctor information and appointment and the relationship between them.
3) View level :- Shows only required information to different users. For example, doctors can see medical history, nurses can see patient vitals, and receptionists can see appointment details.

Thus, data abstraction makes the healthcare system simple, secure and efficient.

Ans:3→ Benefits:—

i) Improves data security and privacy.
ii) Makes the system easy to use for healthcare staff.
iii) Provides data independence, so changes in database structure do not affect users.
iv) Helps in better data management.
v) Challenges:—
i) Designing abstraction layers can be complex
ii) It may cause slight performance overhead.
iii) Implementation can be costly.
iv) Staff may require training to use the system.

Ans:4→ In a university, the main entities are student, course and professor. These entities store important information about student, courses and teachers.

<u>Entities</u> & <u>Attributes</u>:—

i) Student
* student-ID (Primary Key)
* Name
* Gender
* Age
* Email

ii) Course
* Course ID (Primary Key)
* Course - Name.
* Credits
* Department

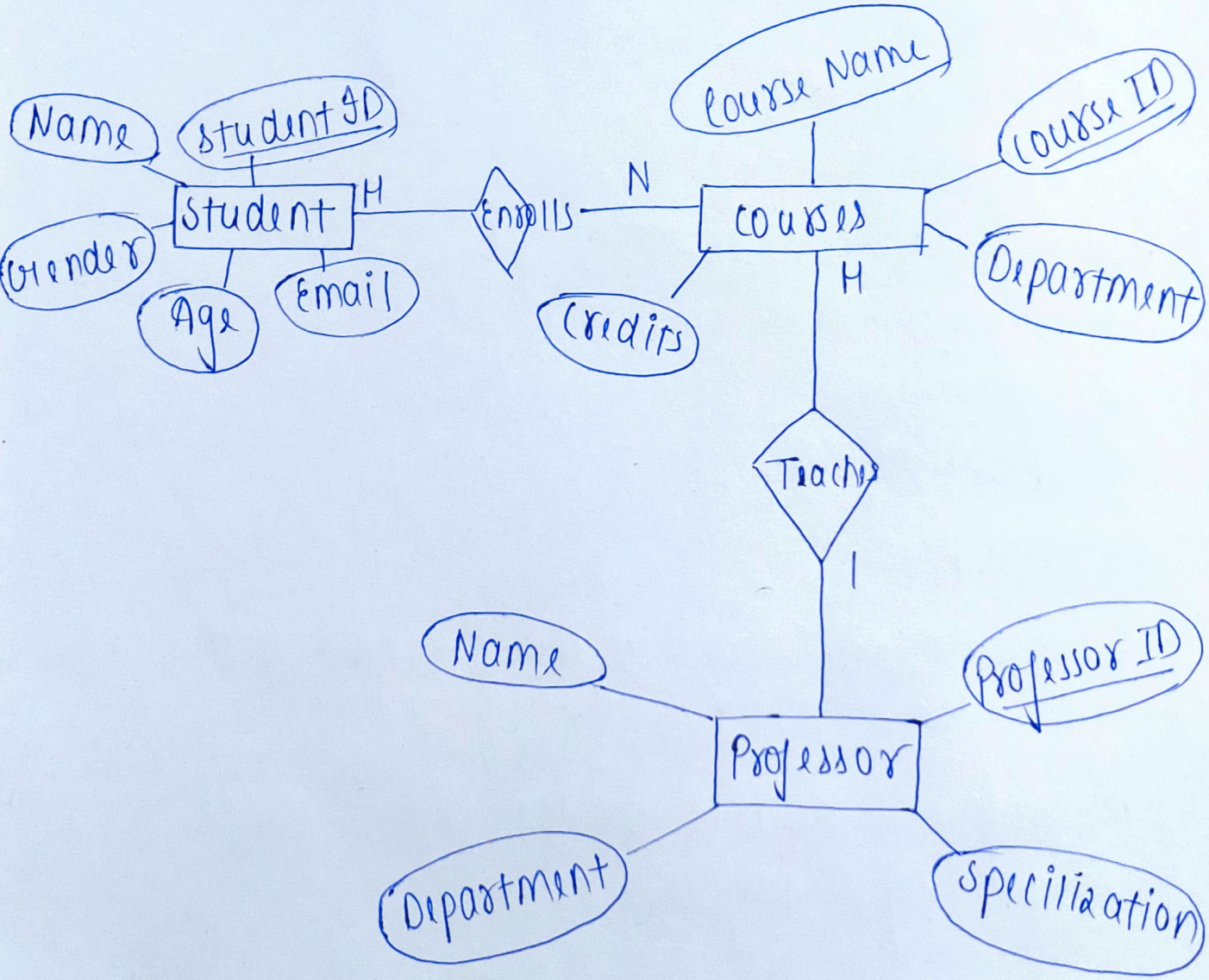
iii) Professor:-
* Professor - ID (Primary Key)
* Professor - Name
* Department
* Specialization

Relationships:-

i) A student can enroll in many courses and course can have ~~s~~ many student.

ii) A professor teaches courses and a course is taught by a professor.

This is one - to many (1:M) relationship.


Name
Student ID
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